



LoC

The logic of Concept definitions

(HP2T)

LoC – The Logic of Concept definitions

- **Intuition**
- Definition
- Domain
- Language
- Interpretation function
- Theories
 - Ontologies, LTLO
- Entailment
- Reasoning problems
 - Satisfiability
 - Subsumption
 - Equivalence
 - Disjointness
 - Reduction to Satisfiability
- Key notions

LoC – Why a logic of Concept definitions?

- The **Logic of concept definitions** (LoC) LoC is a world logic enabling to reason about language models. It allows us to **represent the meanings of concepts** and how their (set-theoretic) meanings correlate. It encodes the meaning of language words,, as represented into ontologies, and used in LoE, LoD and in ALL the other logics, **world logics and beyond**.
- It allows to **reason** about the relations between the meaning of words. For instance it allows to derive that a dog is a living organism but not a person.
- Because of its reasoning capabilities, it allows to represent compactly and to **use adaptively** the meaning implicitly encoded in language words.

Observation (Why LoC). The number of concepts used in natural languages is around 100K, while the concepts used in domain languages (e.g., health) is in the order of hundreds of thousands of concepts. By LoD reasoning it is possible to adaptively generate the minimal but complete concept graph which encodes the meaning of words.

LoC – Highlights

- LoC is the logic encoding the meaning encoded in natural language words, specifically: nouns, verbs, adjectives, adverbs, but NOT names (of entities). It allows to have a precise **unambiguous representation** of the meaning of words, but **not** of the sentences built by using them, which requires using at least LoD.
- All the other world models and world logics must refer to and rely on a specific LoC logic. Whenever this is not done, the resulting world models are incomplete formalizations of the world as they miss all the information implicitly encoded in language.
- LoC allows to encode all types of language models and, therefore, in particular: ontologies, domain languages, domain LTLOs, task LTLOs, application LTLOs.

Observation (why Loc?). The use of LoC in the understanding of the knowledge encoded in a informal or semi-formal world modes requires a preliminary Natural Language step, so called **Word Sense Disambiguation (WSD)**, which “guesses” the meaning of polysemous words (a so called “Informal-to-formal” task).

LoC – which percepts

In LoC we have the following LoC elements:

- A **Concept** is a class of percepts, restricted to the percepts of nouns, verbs, adjectives and adverbs, where ...
- ... the elements of a concept (not represented in LoC and Concept Graphs) can be single elements or tuples of any arity

Example (arity of the elements of a concept): the concept “house” denotes single elements, the concepts “near”, “eat”, “friend” 2-tuples, the concept “between”, 3-tuples, and so on.

Well-formedness conditions

A Concept Graph CG, to be well-formed must satisfy the following conditions:

- Each node is associated one and ONLY one concept.
- Nodes are connected into a tree-like structure
- There is a root. In ontologies the root is usually called “Entity” or “Thing” (T in LoD). In LTLOs, the world labeling the root node depends on the specific purpose and type of the LTLO.

LoC – which alphabet elements?

A Concept Graph CG contains the following elements

- Unique identifiers representing concepts (e.g., words)
- ISA logical symbol used to encode hierarchies representing the fact that the children nodes have extensions which are subsets of the extension of the parent node, and so on down the hierarchy
- Disjointness logical symbol used to encode relations between siblings, where disjointness propagates down to the node children

Observation (ISA, Disjointness). ISA encodes the sharing of meaning across related concepts (as in “person” and “woman”). Disjointness allows to distinguish across unrelated concepts (as in “person” and “animal”, which have disjoint extensions, while being at the same time related, being both “organisms”).

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LoC – The Logic of concept definitions - definition

We formally define LoD as follows

$$\mathbf{LoC} = \langle \mathbf{CG}, \models_{\mathbf{LoC}} \rangle$$

with

$$\mathbf{CG} = \langle \mathbf{L}_{\mathbf{LoC}}, \mathbf{D}, \mathbf{I}_{\mathbf{LoC}} \rangle$$

When no confusion arises, we drop the subscripts.

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Domain

Definition (Domain)

$$D = \langle \{C\} \rangle$$

where:

- $\{u\}$ is a set of **units** $u_1, \dots, u_n$, where $u_1 \in U$, with $U = \{u\}$ is the **universe** of D
- $\{C\}$ is a set of **classes** $C_1, \dots, C_m$ of units, for some m , with $C_i \subseteq U$

Observation (Domain of interpretation). The units in U are not part of the domain of interpretation of LoD. The alphabet does not mention them.

An example of domain of LoC (continued)

$$E_C = \{\text{Entity, P, D, L, ...}\}$$
$$D_C = \{\text{Data, Real, String, ...}\}$$

with, e.g., entity and Real being disjoint domains and so on with their children

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World models as concept graphs (Recall)

Concept graphs are linguistic representations of analogical representations which are used to **define** the meaning of concepts. They are **graphs** where

- nodes are **concepts** (as in «woman», «man», «person»)
 - there are two types of links:
 - «**concept inclusion**» (as in «a woman is a person») and
 - «**concept exclusion**» («**concept disjointness**») (as in «a woman is not a man»)
- relations between two concepts

Reminder (concepts and percepts). Concepts denote sets of percepts... with the possibility of:

- two (sets of) percepts, as denoted by concepts, to be in different places at the same time
- two (sets of) percepts, as denoted by concepts, to be perceived at different times
- two (sets of) percepts, as denoted by concepts, to be perceived in an arbitrarily chosen (but fixed) period of time

Concept graphs (Recall)

Concept graphs take the form of hierarchies where, semantically

- The root node is a concept which denotes the **domain of interpretation**
- A child node denotes a subset of the extension of the parent node (that is, we have **concept inclusion**),
- Any two sibling nodes (i.e., children of the same parent node) are pairwise disjoint (that is, we have **concept exclusion**)

The language of concept definitions

Definition (Language of concept definitions, L_C)

$$L_C = \langle A_C, FR_C \rangle = \{a_{LoC}\}$$

where:

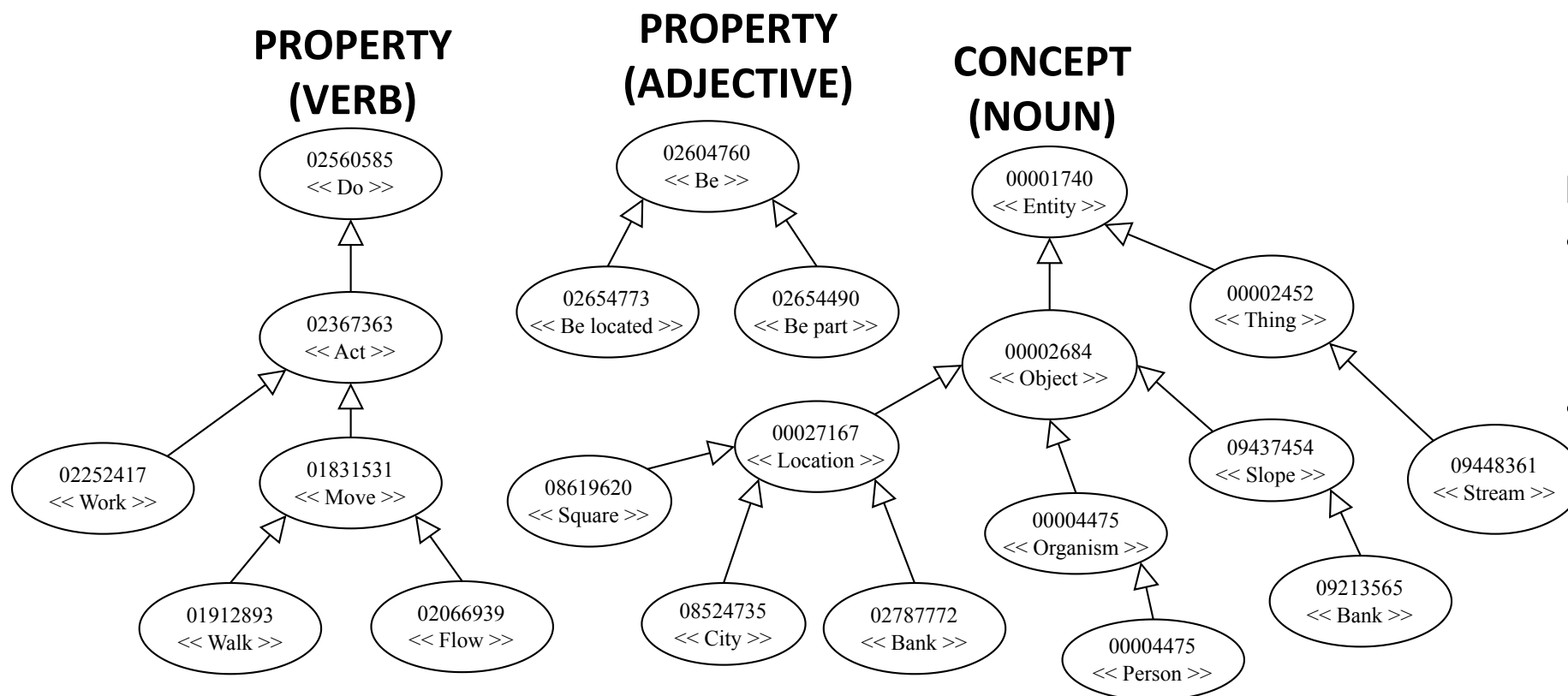
- L_C is a **language** for naming concepts (a **concept language**)
- $A_C = \{c\}$ is an **alphabet of concepts**
- FR_C is a **set of formation rules**
- $\{a_{LoC}\}$ is the set of LoC assertions, i.e., **concept definitions** obtained by the exhaustive application of FR_C to the **concepts** of A_C (the transitive closure $FR_C^+(A_C)$ of FR_C applied to A_C).

Formation Rules – BNF

$$\begin{aligned} \langle a_{\text{LoC}} \rangle \quad ::= & \quad c_1 \sqsubseteq c_2 \mid \\ & \quad c_1 \equiv c_2 \mid \\ & \quad c_1 \perp c_2 \mid \\ & \quad c_1 \perp\!\!\!\perp c_2 \end{aligned}$$

with $c_1, \dots, c_n$ being LoC concepts, that is $c_1, \dots, c_n \in A_C = \{c\}$

Concept graphs – example (Recall)



NOTE:

- All siblings are pairwise disjoint, e.g.,
 $I(\text{organism}) \cap I(\text{location}) = \emptyset$
- All children are subsets of the parents, e.g.,
 $I(\text{city}) \subseteq I(\text{location})$

Formation Rules – BNF

Terminology (Concept subsumption). $\sqsubseteq$ is a **subsumption relation**. $c_1 \sqsubseteq c_2$ is to be read as c_1 is **subsumed by** c_2 , or, vice versa that c_2 **subsumes** c_1 .

Terminology (Concept equivalence). $\equiv$ is an **equivalence relation**. We have

$$c_1 \equiv c_2 \text{ if and only if } c_1 \sqsubseteq c_2 \text{ and } c_2 \sqsubseteq c_1$$

Terminology (concept disjointness): $\perp$ is a **disjointness relation**. $c_1 \perp c_2$ is to be read as c_1 is **disjoint from by** c_2 , and that c_2 is **disjoint with** c_1 . Take $\neg c$ to be the concept such that the union of the intersection of c and $\neg c$ is the universe of interpretation. Then we have

$$c_1 \perp c_2 \text{ if and only if } c_1 \sqsubseteq \neg c_2 \text{ and } c_2 \sqsubseteq \neg c_1$$

Terminology (Strong concept disjointness): $\underline{\perp}$ is a **disjointness relation**. $c_1 \underline{\perp} c_2$ is to be read as c_1 is **strongly disjoint from by** c_2 , and that c_2 is **strongly disjoint with** c_1 . We have

$$c_1 \underline{\perp} c_2 \text{ if and only if } c_1 \equiv \neg c_2$$

Formation Rules – BNF

Observation (BNF LoC). This BNF generates assertions which are triples, i.e., graph assertions.

Observation (BNF LoD). In LoD, etypes can be enriched with properties and composed in more complex etypes. In LoC, this is not possible. Any description of a concept must be done in LoD thus generating an etype.

Observation (BNF LoC). In LoC, disjointness is part of the language, while this is not the case in LoD. In LoD disjointness is part of the query language. In LoD disjointness can be constructed with subsumption and negation but it is not core to the definition of etypes.

Observation (BNF LoC, synonyms, antonyms). In LoC, equivalence and strong disjointness are the key constructs for defining **synonyms** and **antonyms**.

Informal concept definitions – Examples

1. Cars are vehicles
2. Cars are automobiles
3. Humans are not vehicles
4. Humans are not vehicles
5. Vehicles are not humans
6. Women are not men (and vice versa)
7. Anything is an entity
8. An entity is not “not itself”

Formalized concept definitions – Examples

1. $\text{Car} \sqsubseteq \text{Vehicle}$
2. $\text{Car} \equiv \text{Automobile}$
3. $\text{Human} \perp \text{Vehicle}$
4. $\text{Human} \sqsubseteq \neg \text{Vehicle}$
5. $\text{Human} \perp \text{Vehicle}$
6. $\text{Woman} \perp \text{Man}$
7. $C \sqsubseteq \text{Entity}$, for all concepts C
8. $C \perp \neg C$, for all concepts C

NOTE: *same Natural language, different formalization because of the intended model*

LoC assertions – as from the BNF

Observation. LoC allows for the following four assertions:

- $c_1 \sqsubseteq c_2$, a concept is subsumed by another concept
- $c_1 \equiv c_2$, a concept is equivalent to another concept
- $c_1 \perp c_2$, a concept is disjoint with another concept
- $c_1 \perp\!\!\!\perp c_2$, a concept is strongly disjoint with another concept

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Interpretation function

Definition (Interpretation function, alphabet). The **LoC interpretation function** I is defined as

$$I : L \rightarrow D$$

where I is defined as

$$I = \langle I_c \rangle$$

Where $I = \langle I_c \rangle$, the Concept interpretation function, defines the extension of concepts

Interpretation of LoC definitions

$$I(c_1 \sqsubseteq c_2) \quad \text{iff} \quad I_c(c_1) \subseteq I_c(c_2)$$

$$I(c_1 \equiv c_2) \quad \text{iff} \quad I_c(c_1) = I_c(c_2)$$

$$\text{iff} \quad I_c(c_1) \subseteq I_c(c_2) \text{ and } I_c(c_2) \subseteq I_c(c_1)$$

$$I(c_1 \perp c_2) \quad \text{iff} \quad I_c(c_1) \cap I_c(c_2) \subseteq \emptyset$$

$$I(c_1 \perp\!\!\!\perp c_2) \quad \text{iff} \quad I_c(c_1) \cap I_c(c_2) \subseteq \emptyset \text{ and } I(c_1) \cup I(c_2) = U$$

with $c_1, c_2 \in A_C = \{c\}$

LoC Facts

Observation. LoC allows for the following four facts:

- $c_1 \subseteq c_2$, meaning that the extension of a concept is a subset of the extension of another concept.
- $c_1 = c_2$, meaning that the extension of a concept is the same as that of another concept.
- $c_1 \cap c_2 \subseteq \emptyset$, meaning that the extension of a concept is disjoint with that of another concept.
- $c_1 \cap c_2 \subseteq \emptyset$ and $c_1 \cup c_2 = U$, meaning that the extension of a concept is disjoint with that of another concept and their union is the universe of interpretation.

An example of domain of LoC (continued)

We have the following CG

$$E_C = \{\text{Entity, P, D, L, ...}\}$$

$$D_C = \{\text{Data, Real, String, ...}\}$$

from which we construct the following facts in the domain:

$$D = \{P \subseteq \text{entity, Real} \subseteq \text{data, ...}\}$$

with, e.g., entity and Real being disjoint domains and so on
with their children

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LoC theories

Definition (LoC theory). A **LoC theory** $T_{LoC} = \{a\} \subseteq L_{LoC}$ is a set of LoC assertions, where a LoC assertion a has one of four possible forms:

- $c_1 \sqsubseteq c_2$ - A concept is subsumed by another concept
- $c_1 \equiv c_2$ - A concept is equivalent to another concept
- $c_1 \perp c_2$ - A concept is disjoint with another concept
- $c_1 \perp\!\!\!\perp c_2$ - A concept is strongly disjoint with another concept

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Concept graphs (Recall)

Terminology (Language models). Concept graphs are also called **language models** as they unambiguously **define** the meaning of the terms used in (natural / formal) language.

Observation (Language models). Language models allow to create a one-to-one mapping between what we perceive (our analogical representations) and the words (i.e., nouns, verbs, adjectives and adverbs) we use to describe what we perceive.

Terminology (Ontologies). Concept graphs which describe the world as it is / appears are called (in the Artificial Intelligence literature) **ontologies**. They are also called **top level ontologies**.

Terminology (Isa, partof). In ontologies, concept inclusion links are often called **isa (is-a)** links, as in «woman isa person». Some ontologies also have **partof (part-of)** links (as in «leg part-of body») – not considered in this course.

Observation (Language teleontologies, LTLO, domain languages). It is possible to build concept graphs focused on **specific** aspects of the world. We call this type of concept graphs, **language teleontologies (LTLO)**. This leads to the construction of so-called of **domain LTLO** (used in domains such as, e.g., health, spacetime), **task LTLO** (used in tasks such as, e.g., industrial assembly), **application LTLO** (within a specific domain, e.g., health used in specific applications, e.g., hearth surgery). Given their importance and extensive use, domain LTLOs are also called **domain languages**.

LoC terminologies

Terminology (LoC terminologies). LoC theories are called (LoC) terminologies

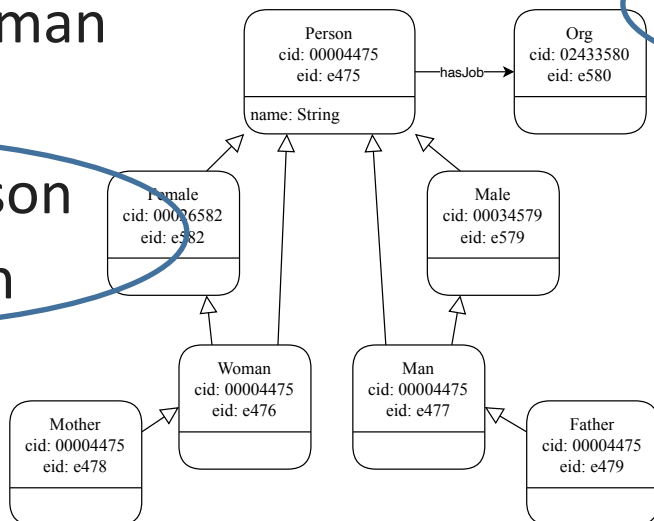
Observation (LoD vs. LoC terminologies). LoC theories are sets of definitions, similarly to LoD terminologies.

Definition (LoC terminologies, ontologies, LTLOs). LoC terminologies are terminologies which define ontologies or LTLOs.

Two KTLOs (example) – DAGs rooted in 1 etype (Recall)

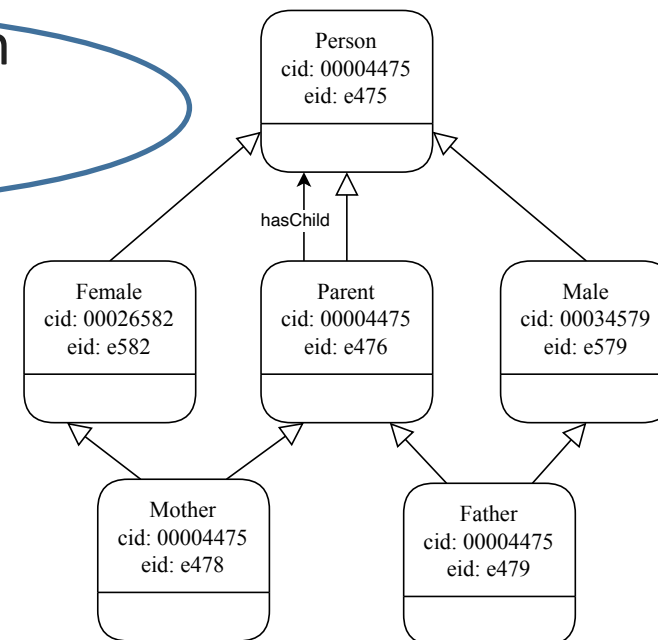
Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$
- $\text{Female} \sqsubseteq \text{Person}$
- $\text{Male} \sqsubseteq \text{Person}$



Family relations

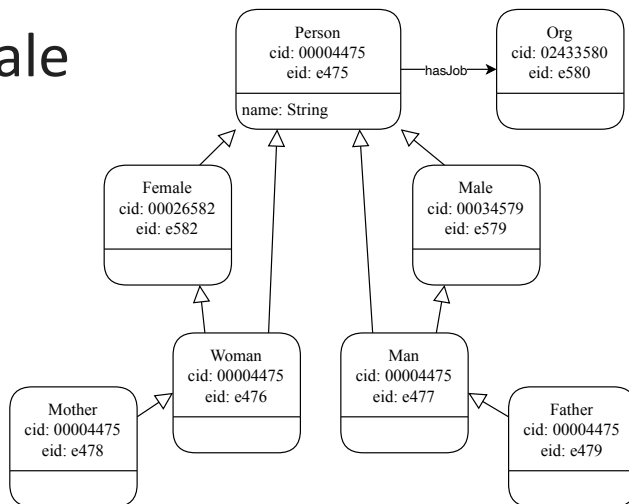
- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild.Person}$
- $\text{Mother} \equiv \text{Parent} \sqcap \text{Female}$
- $\text{Father} \equiv \text{Parent} \sqcap \text{Male}$
- $\text{Female} \sqsubseteq \text{Person}$
- $\text{Male} \sqsubseteq \text{Person}$



From a KTLO DAG to LTLO tree rooted in 1 etype (example)

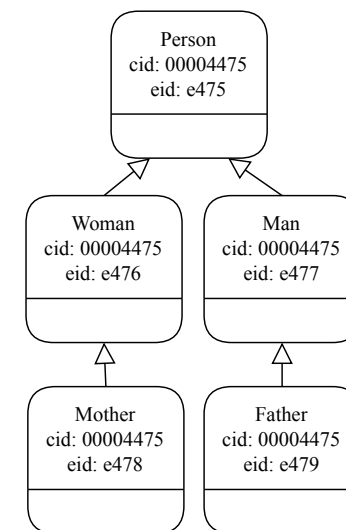
LoD KTLO for family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$
- $\text{Female} \sqsubseteq \text{Person}$
- $\text{Male} \sqsubseteq \text{Person}$



LoC LTLO for family relations

- $\text{Woman} \sqsubseteq \text{Person}$
- $\text{Man} \sqsubseteq \text{Person}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$



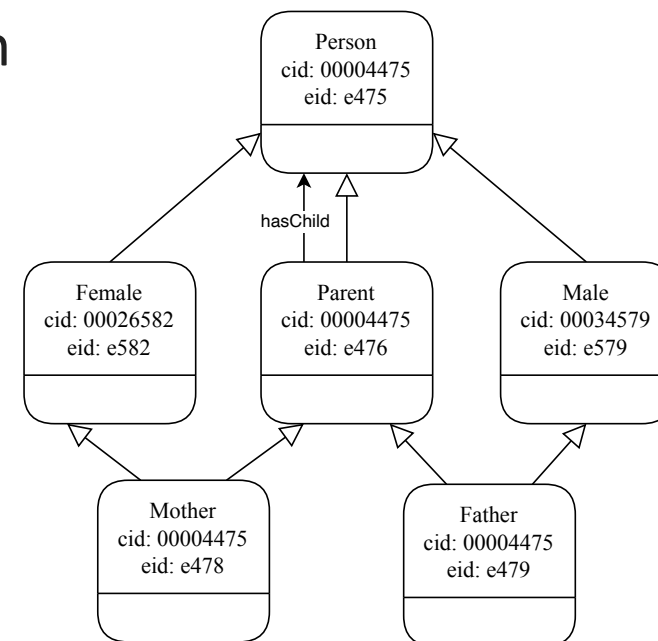
Observation. LoC LTLOs are the backbone of LoD KTLOs, if built well.

Two teleologies (example) – trees rooted in 1 etype

LoD KTLO for family relations

- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild}.\text{Person}$
- $\text{Mother} \equiv \text{Parent} \sqcap \text{Female}$
- $\text{Father} \equiv \text{Parent} \sqcap \text{Male}$
- $\text{Female} \sqsubseteq \text{Person}$
- $\text{Male} \sqsubseteq \text{Person}$

Observation. This LoD KTLO cannot be built from a LoC LTLO. Siblings are not disjoint.



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World entailment (Recall)

Definition (World entailment, satisfiability) Let $W = \langle L_a, D, I_a \rangle$ be a world model. Let L_a be an assertional language. Then $\models_{L_a}$ is a **world entailment relation**

$$\models_{L_a} \subseteq D \times L_a$$

such that

$$M \models_{L_a} T_a$$

$M = \{f\} \subseteq D$ is a **model of** $T_a \subseteq L_a$. We say that M **entails** T_a . We write $M \models T_a$ instead of $M \models_{L_a} T_a$ when no confusion arises.

LoC World Entailment

$$M \models c_1 \sqsubseteq c_2 \quad \text{iff} \quad I_c(c_1) \subseteq I_c(c_2)$$

$$M \models c_1 \equiv c_2 \quad \text{iff} \quad I_c(c_1) = I_c(c_2)$$

$$\text{iff} \quad I_c(c_1) \subseteq I_c(c_2) \quad \text{and} \quad I_c(c_2) \subseteq I_c(c_1)$$

$$M \models c_1 \perp c_2 \quad \text{iff} \quad I_c(c_1) \cap I_c(c_2) \subseteq \emptyset$$

$$M \models c_1 \perp\!\!\!\perp c_2 \quad \text{iff} \quad I_c(c_1) \cap I_c(c_2) \subseteq \emptyset \quad \text{and} \quad I(c_1) \cup I(c_2) \subseteq U$$

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Logical entailment (Recall)

Definition (Logical entailment). Let $L_W = \langle W, \models_W \rangle$ be a world logic with $W = \langle L, D, I \rangle$ its world model. Let $\{M\} \subseteq D$ be a set of models of T with $T, \underline{I} \subseteq L$ and $a \in L$. Then we write

$$T \models_{\{M\}} \underline{I} \qquad (T \models_{\{M\}} a)$$

and say that T **logically entails** $\underline{I}$ (T **logically entails** a) with respect to $\{M\}$ if

$$\text{for all } M \in \{M\}, \text{ if } M \models T \text{ then } M \models \underline{I} \text{ (} M \models a \text{)}$$

Observation (Model filtering via logical entailment). Logical entailment implements reasoning in the following way: “Whatever we derive from T must be true in the same sets of “relevant” models $\{M\}$ as T . T acts as a filter which allows to discard all the irrelevant models.

Observation (Model filtering via selection of $\{M\}$). The choice of the models $\{M\}$ of T is arbitrary. And it also amounts to model filtering. It could be the only model of T , all models of T , a model selected because it is the intended model of T , and so on. If $M \not\models T$ then Logical entailment does not apply. T has the key role of selecting which models $M \in \{M\}$ must be considered.

Notation ($\{M\}$). When no confusion arises we write $\models$ instead of $\models_{\{M\}}$.

LoC – Reasoning problems

Observation (LoC reasoning problems). The four LoC core reasoning problems are:

- $T \models C$ Satisfiability with respect to a TBox T
- $T \models C \sqsubseteq D$ Subsumption with respect to a TBox T
- $T \models C \equiv D$ Equivalence with respect to a TBox T
- $T \models C \perp D$ Disjointness with respect to a TBox T

where T can also be empty.

Observation (LoC vs. LoD). LoC reasoning problems are the same as LoD's, but applied to concepts and not etypes with a language of much lower expressivity

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Satisfiability

 $T \models C$

Satisfiability with respect to a TBox T ,
where C is a concept

Definition (Satisfiability with respect to a TBox T). A concept C is **satisfiable** with respect to a TBox T if there exists a model I of T such that $I(C)$ is not empty. If it is not satisfiable, then C is said to be **unsatisfiable**.

Observation (How to decide Satisfiability). All concepts are satisfiable. There is no way to generate an empty concept.

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Subsumption

$T \models C \sqsubseteq D$ Subsumption with respect to a TBox T

Definition (Subsumption with respect to a TBox T). An assertion C is subsumed by an assertion D with respect to a TBox T if

$$I(C) \subseteq I(D)$$

for every model I of T .

Observation (How to decide subsumption). A concept in a subsumption path is subsubsumed by the concepts above. This can be decided in linear time on the depth of the LoC concept graph.

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Equivalence

$$T \models C \equiv D$$

Equivalence with respect to a TBox T

Definition(Equivalence with respect to a TBox T). Two assertions C and D are equivalent with respect to a TBox T if

$$I(C) = I(D)$$

for every model I of T.

Observation (How to decide disjointness). A concept is disjoint with any type which is not on its subsumption path (from the root). This can be decided in unit time.

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- Entailment
- **Reasoning problems**
 - Satisfiability
 - Subsumption
 - Equivalence
 - **Disjointness**
 - Reduction to Satisfiability
- Key notions

Disjointness

$$T \models C \perp D$$

Disjointness with respect to a TBox T

Definition (Disjointness with respect to a TBox T). Two assertions C and D are disjoint with respect to T if

$$I(C) \cap I(D) = \emptyset$$

for every model I of T .

Observation (How to decide disjointness). A concept is disjoint with any concept which is not on its subsumption path (from the root). This can be decided in linear time on the depth of the LoC concept graph.

LoC – The Logic of Concept definitions

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Reasoning problems (Reduction)

Proposition (Reduction to satisfiability). All the problems reduce to satisfiability. In fact, we have the following equivalences:

- **Equivalence:** $C \equiv_T D$ if and only if $C \sqsubseteq_T D$ and $D \sqsubseteq_T C$;
- **Subsumption:** $C \sqsubseteq_T D$ if and only if $C \sqcap \neg D$ is *unsatisfiable* with respect to T (negation not part of the language. Need to define etype from concept);
- **Disjointness:** $C \perp_T D$ if and only if $C \sqcap D$ is *unsatisfiable* with respect to T .

Observation (Reduction to satisfiability). These reductions are not very useful in LoC as reasoning is straightforward (see above).

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Key notions

- LoC as a logic of concepts
- LoC definitions
- LoC facts
- LoC Terminology
- LoC ontology, LTLO
- Satisfiability
- Subsumption
- Equivalence
- Disjointness
- Reasoning problem reducibility



LoC

The logic of Concept definitions

(HP2T)